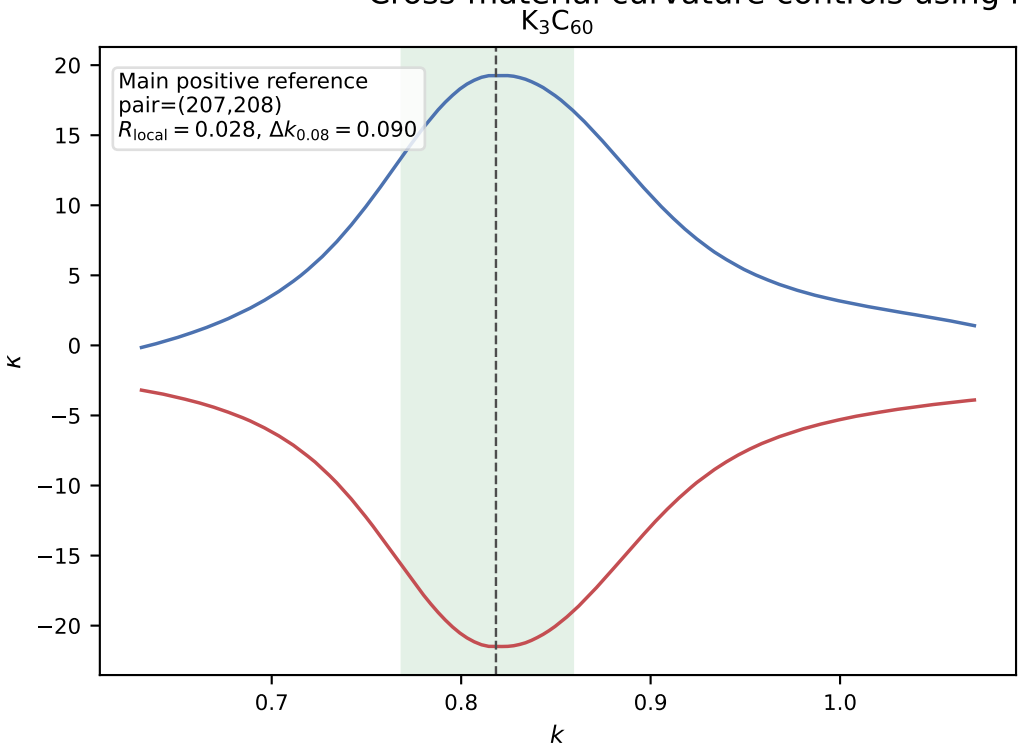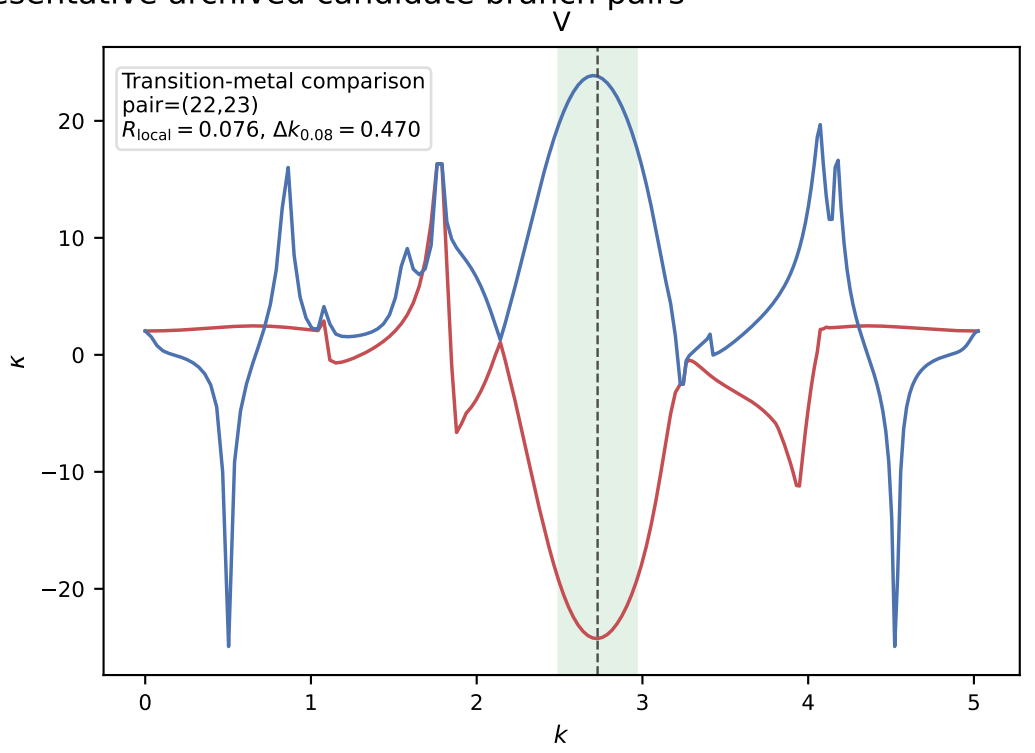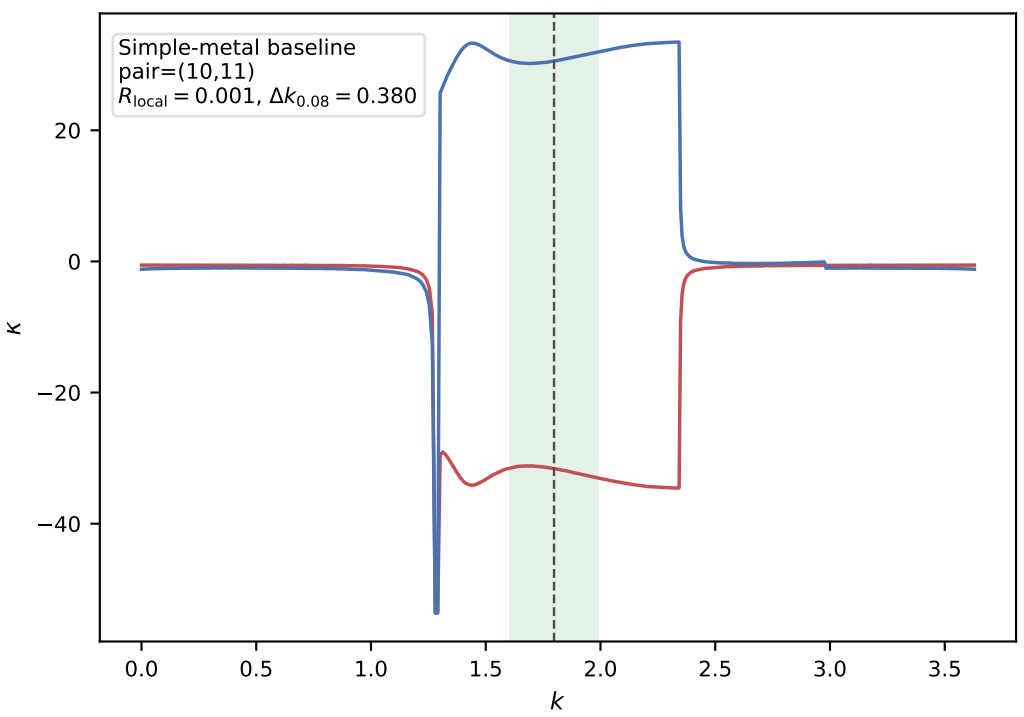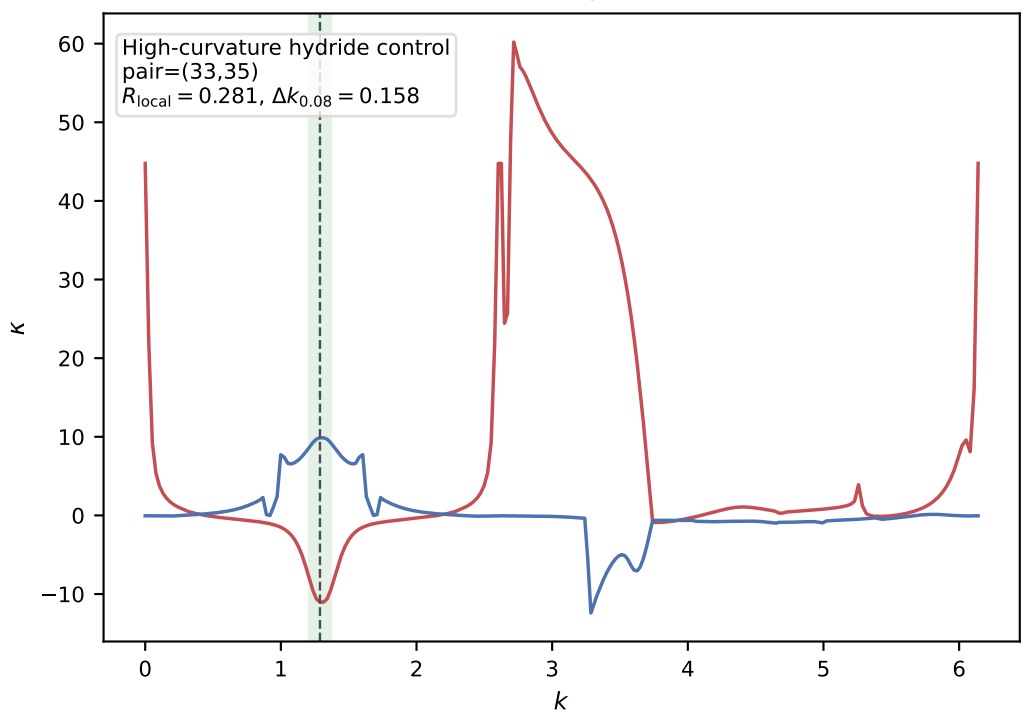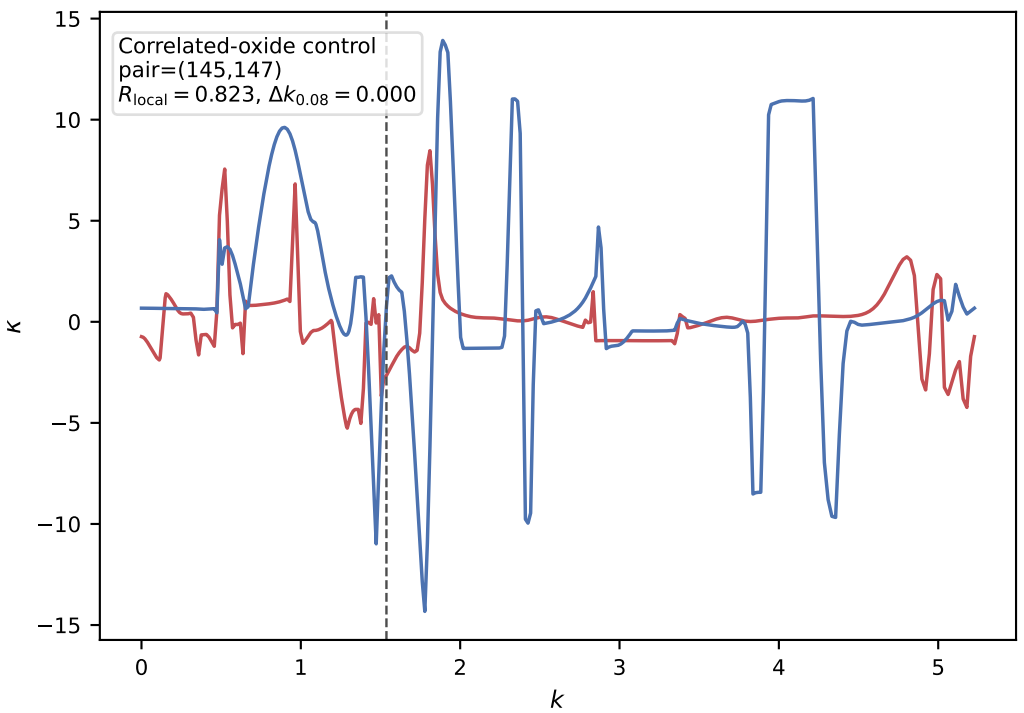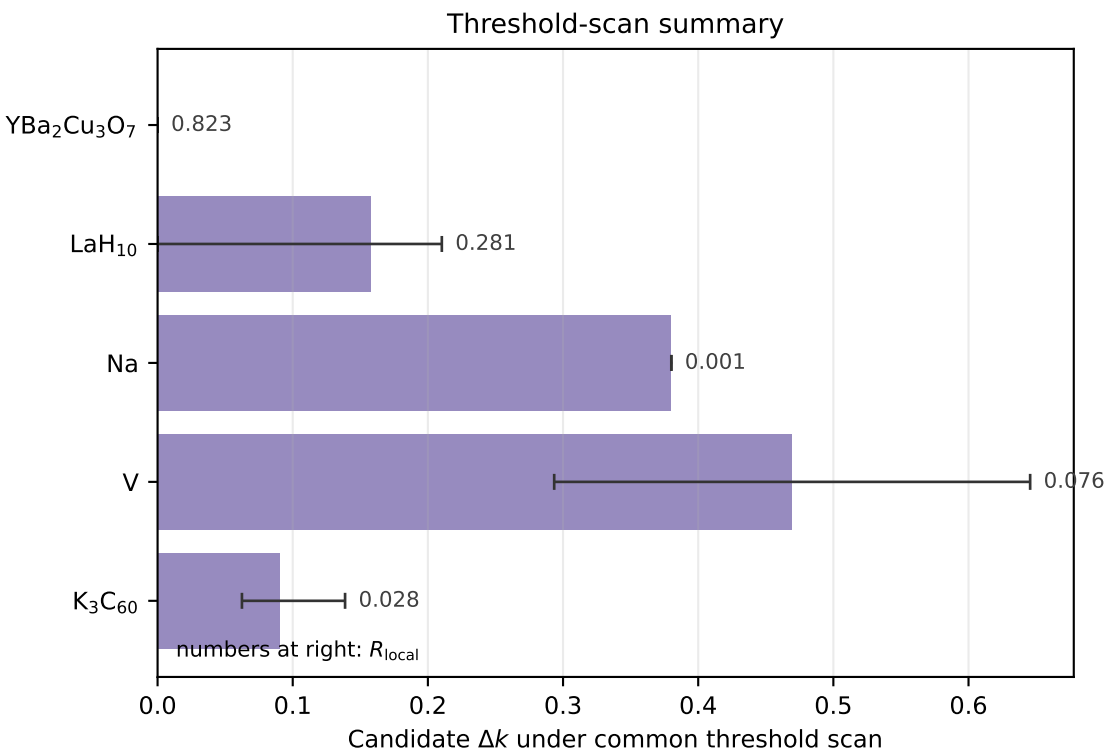Cross-material curvature controls using representative archived candidate branch pairs